

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 10%

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. Detection Performance Comparison

Two object detection models produce the following results:

Model	True Positives	False Positives	False Negatives
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A (Viola-Jones)	80	20	30
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B (YOLO)	90	15	20
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(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

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3. Optical Flow Magnitude Comparison

Optical flow magnitudes (in pixels/frame):

Region	Method A	Method B
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R1	5	8
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R2	6	10
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R3	5	9
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(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

9. Detection Accuracy Comparison

Results:

Model	Correct	Total
A	180	200
B	170	200

- (a) Compute accuracy.
- (b) Compare and evaluate.

10. Error Rate Comparison

Results:

Method	Errors	Total
A	20	100
B	10	100

- (a) Compute error rate.
- (b) Compare and evaluate.

11. Motion Estimation Variance

Errors:

Frame	A	B
1	2	6
2	3	7
3	2	5

- (a) Compute mean error.
- (b) Compare variability and evaluate stability.

13. Recall Comparison in Detection

Results:

Model	TP	FN
A	60	40
B	80	20

- (a) Compute recall.
- (b) Compare and evaluate sensitivity.

14. Depth Estimation Error (Squared)

Predicted:

Object	A	B	Actual
1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

- (a) Compute squared error.
- (b) Compare and evaluate accuracy.

15. Recognition Precision Comparison

Results:

System	TP	FP
A	90	10
B	85	5

(a) Compute precision.

(b) Compare and evaluate performance.

Code of Conduct:

I K.ASWINI(192312397) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison

Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation ; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

ANSWERS

1. Given Data

Model	TP	FP	FN
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Compute Precision and Recall

Formulas

- **Precision** = $TP / (TP + FP)$
- **Recall** = $TP / (TP + FN)$

Model A (Viola-Jones)

- Precision = $80 / (80 + 20) = 80 / 100 = \mathbf{0.80 (80\%)}$
- Recall = $80 / (80 + 30) = 80 / 110 \approx \mathbf{0.73 (72.7\%)}$

Model B (YOLO)

- Precision = $90 / (90 + 15) = 90 / 105 \approx \mathbf{0.86 (85.7\%)}$
- Recall = $90 / (90 + 20) = 90 / 110 \approx \mathbf{0.82 (81.8\%)}$

(b) Performance Comparison

Model	Precision	Recall
A (Viola-Jones)	80%	72.7%
B (YOLO)	85.7%	81.8%

Evaluation

- **Precision:** Model B is higher → fewer false positives
- **Recall:** Model B is higher → fewer missed detections
- Model B performs better in both key metrics

Conclusion

Model B (YOLO) is better for object detection

Justification:

- Higher precision → more accurate detections
- Higher recall → detects more actual objects
- Overall more reliable and robust than Model A

3. Given Data

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) Compute Average Motion Magnitude

Formula

Average = (Sum of values) / (Number of regions)

To visualize how the average is computed (equal distribution of values):

Original amounts
Equal shares

Method A

Average = $(5 + 6 + 5) / 3 = 16 / 3 \approx 5.33$ pixels/frame

Method B

Average = $(8 + 10 + 9) / 3 = 27 / 3 = 9$ pixels/frame

(b) Comparison and Evaluation

- **Method A Average:** 5.33
- **Method B Average:** 9

Analysis:

- Method B shows **higher motion magnitude in all regions**
- Indicates it captures **stronger and more detailed motion**
- Method A shows **lower values** → **smoother but may miss fine motion**

Conclusion

Method B captures stronger motion

Justification:

- Higher average magnitude (9 vs 5.33)
- Consistently detects larger motion across all regions
- More sensitive to motion changes

However, note:

- Higher magnitude may also indicate **more noise sensitivity**
- Method A may still be preferred when **stability is required**

9. Given Data

Model Correct Total

A	180	200
B	170	200

(a) Compute Accuracy

Formula

- **Accuracy = (Correct / Total) × 100**

Model A

$$\text{Accuracy} = (180 / 200) \times 100 = \mathbf{90\%}$$

Model B

$$\text{Accuracy} = (170 / 200) \times 100 = \mathbf{85\%}$$

(b) Comparison and Evaluation

Model Accuracy

A	90%
B	85%

Analysis:

- Model A has **higher accuracy** than Model B
- Difference = **5% improvement**
- Model A makes **fewer errors (20 vs 30 incorrect predictions)**

Conclusion

Model A is better

Justification:

- Higher accuracy (90%) indicates better overall detection performance
- Lower error rate improves reliability in real-world applications

10. Given Data

Frame Method A Method B

1	2	6
2	3	7
3	2	5

(a) Compute Mean Error

Formula

$$\text{Mean} = (\text{Sum of errors}) / \text{Number of frames}$$

Method A

$$\text{Mean} = (2 + 3 + 2) / 3 = 7 / 3 \approx \mathbf{2.33}$$

Method B

$$\text{Mean} = (6 + 7 + 5) / 3 = 18 / 3 = \mathbf{6}$$

(b) Compare Variability and Evaluate Stability

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Method A Variability

- Errors are close: 2, 3, 2
- Small deviation from mean (≈ 2.33)
Low variance $\rightarrow$ High stability

Method B Variability

- Errors: 6, 7, 5
- Slightly larger spread around mean (6)
✓ Higher variance than A $\rightarrow$ Less stable

Comparison

Metric	Method A	Method B
Mean Error	2.33	6
Variability	Low	Moderate
Stability	High	Lower

Conclusion

✓ Method A is more stable and accurate

Justification:

- Lower mean error $\rightarrow$ better accuracy
- Lower variance $\rightarrow$ consistent performance
- More reliable for motion estimation tasks

11. Given Data

Frame Method A Method B

Frame Method A Method B

1	2	6
2	3	7
3	2	5

(a) Mean Error

Mean error is the average of errors across frames.

- **Method A:** $(2 + 3 + 2) / 3 = \mathbf{2.33}$
- **Method B:** $(6 + 7 + 5) / 3 = \mathbf{6.00}$

(b) Variability (Stability Analysis)

To judge stability, compute **variance** (lower = more stable).

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Method A

- Mean = 2.33
- Squared deviations:
 $(2 - 2.33)^2 \approx 0.11$
 $(3 - 2.33)^2 \approx 0.44$
 $(2 - 2.33)^2 \approx 0.11$
- Variance = $(0.11 + 0.44 + 0.11) / 3 \approx \mathbf{0.22}$

Method B

- Mean = 6
- Squared deviations:
 $(6 - 6)^2 = 0$
 $(7 - 6)^2 = 1$
 $(5 - 6)^2 = 1$
- Variance = $(0 + 1 + 1) / 3 \approx \mathbf{0.67}$

Comparison

Metric	Method A	Method B
Mean Error	2.33	6.00
Variance	0.22	0.67
Stability	High	Moderate

Conclusion

✓ **Method A is better**

Justification:

- Lower mean error → more accurate motion estimation
- Lower variance → more consistent (stable) performance
- Method B has higher error and more fluctuation

Final Decision: Choose **Method A** for reliable and stable motion estimation.

13. Given Data

Model TP FN

A 60 40

B 80 20

(a) Compute Recall

Formula

- **Recall = TP / (TP + FN)**

Model A

Recall = $60 / (60 + 40) = 60 / 100 = \mathbf{0.60 (60\%)}$

Model B

Recall = $80 / (80 + 20) = 80 / 100 = \mathbf{0.80 (80\%)}$

(b) Comparison and Evaluation (Sensitivity)

Model Recall

A 60%

B 80%

Analysis:

- Recall measures **sensitivity** (ability to detect actual positives)
- Model B detects **more true positives** and misses fewer cases
- Model A has higher false negatives → misses more objects

Conclusion

Model B is more sensitive and better for detection tasks

Justification:

- Higher recall (80%) → better at identifying actual objects
- Lower miss rate improves performance in critical applications (e.g., surveillance, medical detection)

14. Given Data

Object Method A Method B Actual

1	2.5	2.2	2
2	3.5	3.1	3
3	4.5	4.2	4

(a) Compute Squared Error

Formula

- **Squared Error = (Predicted – Actual)²**

Method A

Object Error Squared Error

1	$2.5 - 2 = 0.5$	0.25
2	$3.5 - 3 = 0.5$	0.25
3	$4.5 - 4 = 0.5$	0.25

Total Squared Error (A) = $0.25 + 0.25 + 0.25 = 0.75$

Mean Squared Error (A) = $0.75 / 3 = 0.25$

Method B

Object Error Squared Error

1	$2.2 - 2 = 0.2$	0.04
2	$3.1 - 3 = 0.1$	0.01
3	$4.2 - 4 = 0.2$	0.04

Total Squared Error (B) = $0.04 + 0.01 + 0.04 = 0.09$

Mean Squared Error (B) = $0.09 / 3 = 0.03$

(b) Comparison and Evaluation

Metric	Method A	Method B
Total Error	0.75	0.09
Mean Squared Error	0.25	0.03

Analysis

- Method B has **much lower squared error**
- Predictions of Method B are **closer to actual values**
- Method A consistently overestimates by 0.5

Conclusion

Method B is more accurate

Justification:

- Lower MSE (0.03 vs 0.25)
- Smaller deviation from actual depth values
- Better reliability in depth estimation tasks

15. Given Data

System TP FP

A 90 10

B 85 5

(a) Compute Precision

Formula

- **Precision = TP / (TP + FP)**

System A

$$\text{Precision} = 90 / (90 + 10) = 90 / 100 = \mathbf{0.90 (90\%)}$$

System B

$$\text{Precision} = 85 / (85 + 5) = 85 / 90 \approx \mathbf{0.944 (94.4\%)}$$

(b) Comparison and Evaluation

System Precision

A	90%
B	94.4%

Analysis

- System B has **higher precision**
- Lower false positives in B (5 vs 10)
- B is more reliable when avoiding incorrect detections is important

Conclusion

System B performs better in terms of precision

Justification:

- Higher precision (94.4%) → fewer false alarms
- More trustworthy predictions in recognition tasks